

COMP 1005B/1405B

Winter 2022 - Tutorial 2

Objectives

- Practice creating **pseudocode**/flowchart algorithms
- Practice using basic Python **print** function
- Practice using basic Python **input/output operations** and **math operators**

Expectations

To receive full grades for this tutorial, your group must complete Problems 1-4. Problem 5 is for practice. If you do not have time to complete the required problems, you should attempt them after the tutorial for your own benefit.

Submit

For this tutorial, your group will submit a single zip file called **YourGroupName_T2.zip** to the appropriate channel in Discord, where **YourGroupName** should be replaced by *your group name*. The zip file should contain the attendance record of your group members and the solutions to all the required problems. If any of your group members is absent during a tutorial session and does not participate in any group work during that entire week, you must report it.

Ensure that your tutorial grade is entered on Brightspace at the end of each tutorial week. If you don't see your previous grades, contact the Tutorial TAs immediately. You have no more than 1 week to report a missing tutorial grade.

You can read about certain built-in python functions that will be used in this tutorial here: <https://docs.python.org/3/library/stdtypes.html>

Warm-up

Problem 1 (Algorithms - pseudocode and flowchart)

Write the pseudocode using a text editor or a word processor. Draw the flowchart either on a paper and scan it, or use any drawing tool. Copy-paste the answers to a file and save it as **p1.pdf**.

- a) Create a pseudocode and a flowchart for a program that asks the user to enter two numbers and print out which of the two numbers is the largest.
- b) Discuss how this pseudocode and flowchart can be modified for a program that asks the user to enter three numbers and print out which of the three numbers is the largest. You do not need to submit the algorithm (pseudocode or flowchart) for this problem.

Problem 2 (Calculations)

Write a program called **p2.py** that implements a tip calculator.

Your program should ask the user for the bill amount and the percentage of tip, and then output the dollar amount of the tip and the final bill in three different formats (see sample below).

Sample output (the user input is highlighted):

```
Enter the bill amount: 27.95
Enter tip percentage: 17
A 17.0% tip on $27.95 equals $4.7515
The total bill is: 32.701499999999996
The total bill is: $32.7
The total bill is: $32.70
```

Notes:

- The `input()` function returns the input as a string. In order to convert the string into an integer use the `int()` function. For example, if `x = "12"`, then `int(x)` is the integer 12.
- Also recall that the `str(number)` function will convert a number to a string. You need to use this to concatenate numbers with the strings (e.g., `str1 + str(number)`) before it can be printed out.
- The 4th line of the output prints the total bill amount in its original format.

- The 5th line of the output prints the dollar amounts rounded to two decimal places (though python will automatically remove trailing zeroes). The function `round(num, [precision])` rounds a given number to the specified precision (default is 0). For example,
`x = round((3.1415926, 4))` gives `x = 3.1416`.
- The 6th line of the output prints the dollar amounts with exactly two decimal places. Several string formatting options are available in Lecture-04. You may also read section 6.1.3 'Format String Syntax' from the official python documentation website <https://docs.python.org/3.4/library/string.html> and see all format options that will print the output with exactly 2 places after the decimal.

Hangman Game

The Hangman Game problems of every tutorial will help you build and solidify your skills learned in this course while working towards developing a working hangman game. All of the problems will be building blocks or pieces of the final game.

If you want to refresh your memory on the general idea of the game, you can read about it here: [https://en.wikipedia.org/wiki/Hangman_\(game\)](https://en.wikipedia.org/wiki/Hangman_(game))

Problem 3 (Let's print the hangman)

Write a program called **p3.py** that prints the following string (hangman stick figure) on the terminal.

```

  o
  \|/
  |
  /\

```

Note: you need to understand '**escape sequences**' to correctly print the backslash. For more information, see

https://www.w3schools.com/python/gloss_python_escape_characters.asp
<http://www.python-ds.com/python-3-escape-sequences>.

Problem 4 (Input / Output)

Write a program called **p4.py** that asks the user to enter their name and a letter. The program should then use this information to output the string “<your name> guessed the letter <lowercase letter>”, where <your name> is the name entered by the user and <lowercase letter> is the lowercase form of the letter entered by the user.

tip: make use of the method `.lower()`

example of the `.lower()` method:

```
>>> x = COMP
>>> print(x.lower())
>>> comp
```

Sample output 1:

```
Enter your name: Jane Doe
Guess a letter: A
Jane Doe guessed the letter a
```

Sample output 2:

```
Enter your name: Jane Doe
Guess a letter: n
Jane Doe guessed the letter n
```

Note: This problem is meant to help you practice using input, print, variables, string concatenation, and python built-in methods.

DNA Project

The DNA Project section of the tutorials contains challenge problems. You are not required to submit answers to these questions, however you will greatly benefit from completing them every week. These problems are meant to be more challenging than the Hangman Game problems. All of the problems in this section are building towards a final DNA Project.

These problems won't require advanced knowledge of DNA or biology, but here are the basics that you might find helpful:

- At a very high level, DNA is made up of two strands of chained nucleobases.
- They are made up of combinations of cytosine (C), guanine (G), adenine (A), and thymine (T).
- DNA strands are in pairs that are complements of each other. A and T are complements and C and G are complements.
- example of a pair: ATTGCC and TAACGG

Note the above information is a very basic, simplified 'version' of DNA (not perfectly accurate).

Problem 5 (Formatting)

Write a program called **p5.py** that takes a string as input and outputs the string in all capital letters and the length of the string. The output should follow the format from the example below.

Sample output:

```
Enter a DNA strand:  aAtc
DNA strand:         AATC
Length:             4
```

Tip: make use of the `.upper()` method, the `len()` function, and the `'\t'` escape character to add tabs.

You can read more about each of these helpful tools here:

`.upper()`: https://www.w3schools.com/python/ref_string_upper.asp

`len()`: https://www.w3schools.com/python/ref_func_len.asp

escape characters: https://www.w3schools.com/python/gloss_python_escape_characters.asp